



Development of a machine learning model to classify mineral deposits using sphalerite chemistry and mineral assemblages

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ABSTRACT

Sphalerite is a commonly occurring mineral in natural systems and a prominent indicator mineral used in resource exploration. Its chemistry and associated mineral assemblages are both controlled by the physico-chemical conditions of the local environment, such as temperature and sulfur fugacity. Accordingly, the chemistry of sphalerite and the nature of the associated minerals provide valuable clues about the classification of mineral deposits and their environment of formation. Nevertheless, exclusive reliance on the trace-element chemistry of sphalerite for deposit classification has its limitations given the multitude of factors (For example, deposit type, temperature, pressure, and background concentrations of elements) that affect its chemistry. To address this challenge, we develop machine learning models using the SHapley Additive exPlanations (SHAP) method to assess the importance of sphalerite trace-element chemistry and mineral assemblage information to distinguishing between five petrogenetically distinct mineral deposit types. This contribution demonstrates that a composite model that incorporates both data types markedly improve the accuracy of deposit type discrimination. The composite model is composed of two sub-models, Random Forest (RF) and Extra Random Trees (ERT), which are specifically employed for processing trace-element data and mineral assemblage data, respectively, due to their superior performance in handling these two distinct types of datasets. To simplify the end-user experience, we provide an executable file of the machine learning-based classifier, allowing it to be readily applied as an exploration tool using simple inputs. In summary, this work substantially enhances the confidence of Pb-Zn deposit type classification using sphalerite and introduces an innovative perspective on the application of machine learning to resolving complex geological problems.

1. Introduction

Indicator minerals are an important tool in mineral resource exploration due to their distinct chemical compositions, and resilience to weathering and erosion, allowing them to serve as physicochemical 'signposts' for different deposit types (McClenaghan, 2005; Averill, 2007; Ferbey et al., 2017). The chemistry, morphology, and assemblages of these minerals provide invaluable clues about the genesis and evolution of mineral deposits, mineralizing processes that led to their formation, and their economic viability (Ridley, 2013). By directing exploration efforts, and reducing the reliance on costly and invasive

exploration methods (e.g., trenching), indicator minerals substantially improve the efficacy of mineral resource exploration strategies (Chapman et al., 2000; Stachel and Harris, 2008).

Trace elements in indicator minerals serve as powerful tools for characterizing deposit types (Liu and Beaudoin, 2021). This is because the concentrations of trace elements can record metallogenic information, such as the temperature, oxygen fugacity, and composition of ore-forming fluids, thereby providing key information about the environment of deposit formation, which is intrinsically linked to deposit class. A variety of minerals have been applied as indicators for mineral deposit classification and exploration, including, but not limited to, quartz

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(Maneta and Baker, 2019; Wang et al., 2021), magnetite (Pisiak et al., 2017; Huang et al., 2019), pyrite (Zhang et al., 2017; Duran et al., 2019), and garnet (Baxter et al., 2017; Ding et al., 2018). In addition to indicator mineral chemistry, the assemblages of minerals that accompany indicator minerals are an essential feature for distinguishing among deposit types. This is because different deposit types comprise different suites of minerals based on the physicochemical conditions of their environment of formation, such as temperature, pressure, fluid chemistry and pH, and oxygen and sulfur fugacities (Guilbert and Park Jr, 2007). This paragenetic information serves as a fingerprint for deposit type identification, and provides clues about the geological history and metallogenic processes that led to deposit formation (Barnes, 1997).

Machine learning (ML) provides an optimal framework to undertake the multivariate analyses required to assess the robustness of mineral trace-element chemistry and mineral assemblage information to mineral deposit classification and prospectivity assessment as this technique can process and evaluate large amounts of high-dimensional data (Ueki et al., 2018; Saha et al., 2021). ML algorithms have been increasingly utilized in geoscience to tackle complex problems, such as predicting mineral formation temperatures (Li and Zhang, 2022), quantifying crustal thickness (Zou et al., 2021; Guo and Yang, 2023), assessing magma fertility (Zou et al., 2022), and distinguishing tectonic environments (Doucet et al., 2022). In the context of ore deposit science, ML has been demonstrated to be a powerful method for deposit-type discrimination. For example, in the past five years, ML models that discriminate among deposit types have been developed using the trace-element chemistry of magnetite (Hong et al., 2021), quartz (Wang et al., 2021), and pyrite (Gregory et al., 2019). Recent applications of ML models to characterizing the genesis of sphalerite-bearing deposits (Sun and Zhou, 2022; Li et al., 2023) has demonstrated the possibility of using sphalerite chemistry to classify, or differentiate between, mineral deposit types. However, evaluation of models' generalization performance are essential to distinguishing different deposit types, and accurate and consistent explanations regarding the contribution and importance of physicochemical features to deposit classification play crucial roles in creating robust ML models.

In this contribution, we aim to address these limitations by developing a composite model based not only on sphalerite chemistry, but

also the mineral assemblages associated with the sphalerite. To accomplish this, we collated a comprehensive dataset of 3159 sphalerite analyses from 109 different deposits around the world and evaluated the generalization performance of two individual models, one based on sphalerite chemistry and the other based on associated mineral assemblages, using random sampling. We then constructed a composite ML model composed of the two individual ML models. Finally, we employed the SHapley Additive exPlanations (SHAP) method to assess the importance of trace-element concentrations and mineral assemblages to deposit classification. It should be additionally noted that we attempted to consider a broader range of deposit types. However, due to sphalerite not being a principal mineral in porphyry or orogenic deposits, and the extremely scarce data on trace elements in sphalerite from these two types of deposits, this led to poor classification results (The classification accuracy for these two deposit types is less than 50 %). Therefore, we only considered the major types of Pb-Zn deposits, including Volcanogenic Massive Sulfide (VMS), Mississippi Valley-Type (MVT), epithermal, Sedimentary Exhalative (SEDEX), and skarn deposits.

2. Data and method

2.1. Data source

Sphalerite trace-element chemistry (obtained primarily via LA-ICP-MS) and associated mineral assemblages from various deposit types around the world were collated from literature (Fig. 1). The dataset comprises 3159 sphalerite analyses from 109 ore deposits that span five deposit types — VMS, MVT, epithermal, SEDEX, and skarn deposits (Supplementary Data Table S1). The trace-element content of sphalerite and the mineralogy of the associated assemblages were selected from each deposit as training features for the ML models, and comprise 12 trace elements (Mn, Fe, Co, Cu, Ga, Ge, Ag, Cd, In, Sn, Sb, and Pb) and 15 minerals (sphalerite, galena, pyrite, chalcopryite, tetrahedrite, tellurides, magnetite, stannite, fluorite, cassiterite, enargite, stibnite, arsenopyrite, tennantite, pyroxene). For mineral assemblage data, we parameterized the presence or absence of each mineral in a given deposit, with presence and absence marked as 1 and 0, respectively.

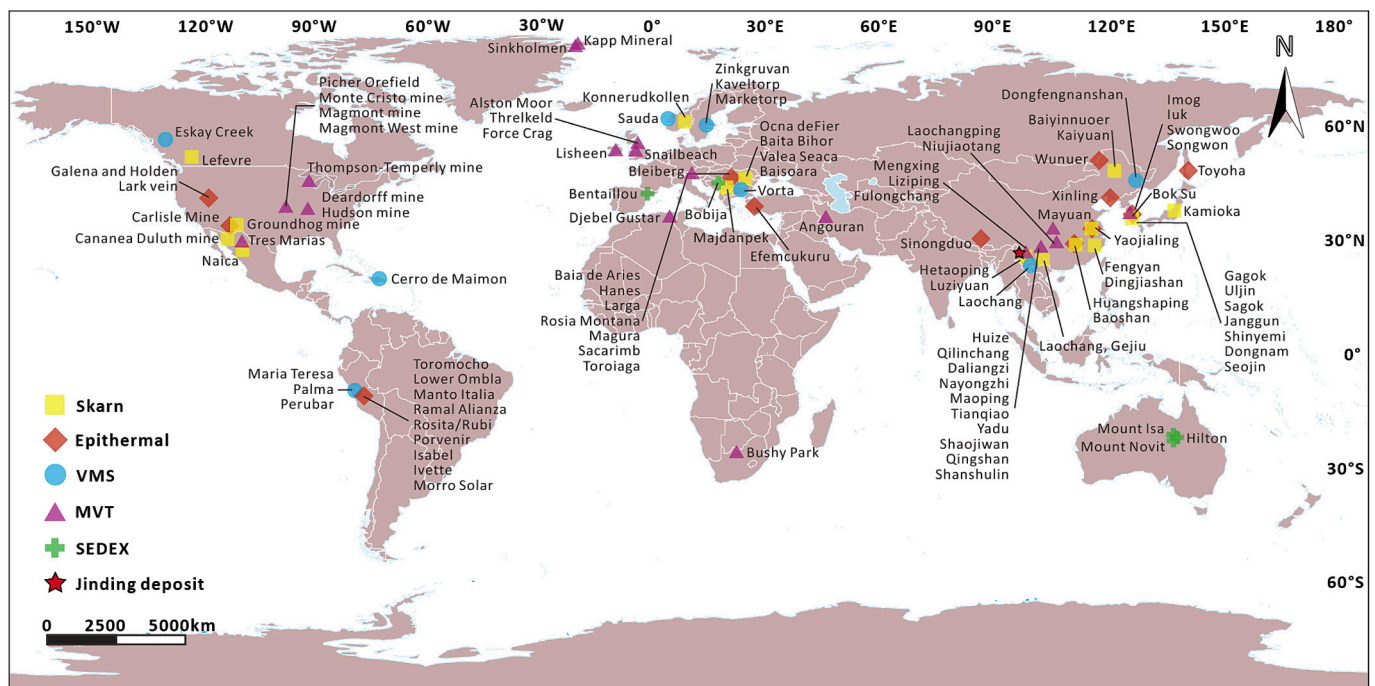


Fig. 1. World map showing the distribution of sphalerite data collected in this study.

2.2. Data preprocessing

Outlier detection and handling is a critical step in data preprocessing for many scientific studies, including in the fields of machine learning and statistics. Various methods exist for identifying and managing outliers, including Z-score, modified Z-score, IQR, and various statistical and machine learning algorithms, such as DBSCAN and Isolation Forest (Chandola et al., 2009; Aggarwal, 2017). While these methods can effectively detect outliers, the decision to remove, replace, or retain outliers should be made carefully and should be dependent on the nature and purpose of the study. Indiscriminate removal of outliers can potentially lead to the loss of important information and may even create bias given that outliers can sometimes highlight important patterns or errors in the data (Barnett and Lewis, 1994; Aggarwal, 2017). If not handled correctly, outliers may skew model parameters and adversely affect the performance of models, particularly those based on assumptions of normally distributed residuals (Hawkins, 1980). Given the complexity of geochemical data, it is not straightforward to visually identify outliers. In light of the suboptimal performance exhibited by the Z-score method (76 % of all entries were identified as outliers using the Z-score method), we employed the IQR method for the purpose of outlier removal.

The handling of below detection limit (BDL) values is a critical step in data preprocessing for statistical analysis and machine learning applications (Helsel, 2005; Thomas, 2006). The common practice of substituting BDL values with zero introduces bias into the dataset since BDL values represent a range and not an absolute zero (Antweiler and Taylor, 2008; Helsel, 2010). Such practices can result in underestimation and distortion of the dataset's distribution, potentially leading to misleading inferences and erroneous conclusions. Moreover, it dismisses the inherent uncertainty associated with BDL values, which is crucial for accurately characterizing the dataset and for maintaining the quality of interpretations (Thomas, 2006). Hence, employing more robust methods for handling BDL values is imperative to mitigating potential biases and adverse effects related to simply substituting BDL values with zero (Helsel, 2005). We addressed BDL values in our compiled dataset by employing random imputation using a normal distribution, with both mean and standard deviation equal to the detection limit, while adjusting for negative values by detection limit values (van den Boogaart and Tolosana-Delgado, 2013). It is crucial to acknowledge that this method may overestimate the mean, especially when numerous BDL values are present, potentially diminishing the observed differences between sample populations.

In this study, we first employed the IQR method to eliminate outliers. Subsequently, BDL values were replaced with normally distributed random numbers, where the mean and standard deviation were set to the detection limit. Finally, missing values were imputed using the multiple imputation method. To mitigate the effects of sample imbalance arising from the natural heterogeneity in the population size of different deposit types, weights were introduced during the model training process. Specifically, weights were assigned to samples based on their deposit type within the training set, with the weight being 1 divided by the number of samples belonging to the respective type. Subsequently, the weight data were input into the model alongside the training set during the model fitting process.

2.3. Machine learning algorithms

Multiple ML algorithms are developed using the sphalerite trace-element chemistry and the associated mineral assemblage dataset. Ensemble methods, artificial neural networks (ANNs), and support vector machines (SVMs) are commonly used ML algorithms due to their efficacy in managing complex classification and regression tasks (Kotsiantis et al., 2007). Extremely Randomized Trees (ERT) and Random Forest (RF) are ensemble methods that construct multiple decision trees, with the final prediction determined by aggregating individual tree

outputs through voting or averaging (Breiman, 2001; Geurts et al., 2006). Both algorithms, however, can become computationally expensive and less interpretable as the number of trees increases (Hastie et al., 2009). Similarly, Gradient Boosting Decision Trees (GBDT) and XGBoost iteratively construct decision trees, focusing on reducing the residual errors of previous trees using gradient descent optimization, but are prone to overfitting and require careful tuning of parameters, such as learning rate and tree complexity (Friedman, 2001; Chen and Guestrin, 2016). ANNs emulate the biological structure and function of the brain's neural networks, leveraging interconnected nodes (neurons) organized into layers to learn patterns from input data for making predictions, but are notorious for their "black-box" nature and difficulty in interpretation, and are susceptible to local minima during optimization (Goodfellow et al., 2016). SVMs, on the other hand, represent a class of supervised learning algorithms that seek the optimal hyperplane separating the classes in the feature space, thereby maximizing the margin between the classes. SVMs can also handle nonlinear classification through the use of kernel functions (Cortes and Vapnik, 1995). Although SVM are effective in high-dimensional spaces, they can struggle with large datasets and provide limited interpretability due to their non-probabilistic nature (Cortes and Vapnik, 1995).

Hyperparameter tuning was performed in two steps. Initially, a relatively narrow range of hyperparameters were determined using Grid Search (a hyperparameter tuning method provided by the Scikit-learn Python library). This range was subsequently fine-tuned manually to improve precision. All computations were implemented in the Python programming language.

3. Model construction

3.1. Strategies

Fig. 2 illustrates the workflow utilized in the development of the various machine learning models. Important aspects of this workflow are described below.

a. Optimal algorithm selection: Even within the same deposit type, there exist variations in trace elements in sphalerite, such as in VMS deposits (Supplementary Data Fig. S1). This variability poses a challenge for the model's classification tasks. In practical applications, we aim to predict entirely unknown deposits, thus, deposits that appear in the training set must not occur in the test set. Otherwise, it would lead to severe overfitting (Sun and Zhou, 2022). To effectively evaluate the generalization performance of the different algorithms and prevent overfitting, training and test datasets were constructed at the deposit level. To accomplish this, we randomly selected samples from 20 % of the dataset to be used as the test set and the remaining 80 % of samples were used as the training set. To minimize the impact of chance errors, we compared the classification performance of each algorithm based on 100 iterations. Importantly, these assessments were performed separately on the sphalerite trace-element data and the associated mineral assemblage data.

b. Feature importance: SHAP is a unified measure of feature importance that assigns each feature an importance value for a particular prediction (Lundberg and Lee, 2017). The concept is based on Shapley values, which, in cooperative game theory, is used to fairly distribute the payout among the players depending on their contribution to the total payout (Shapley, 1953). In the context of machine learning, each feature is considered a player that collaborates to predict the target output. The SHAP value of a feature is the average marginal contribution of the feature across all possible feature subsets (Lundberg et al., 2020). Importantly, not only do SHAP values provide a global measure of feature importance, but also offer local interpretability by explaining the contribution of each feature to every individual prediction. This is particularly beneficial in explaining the feature importance of individual categories. Overall, its unique advantages include: 1) Consistency: Ensures stable and reliable importance values as models change. 2) Local

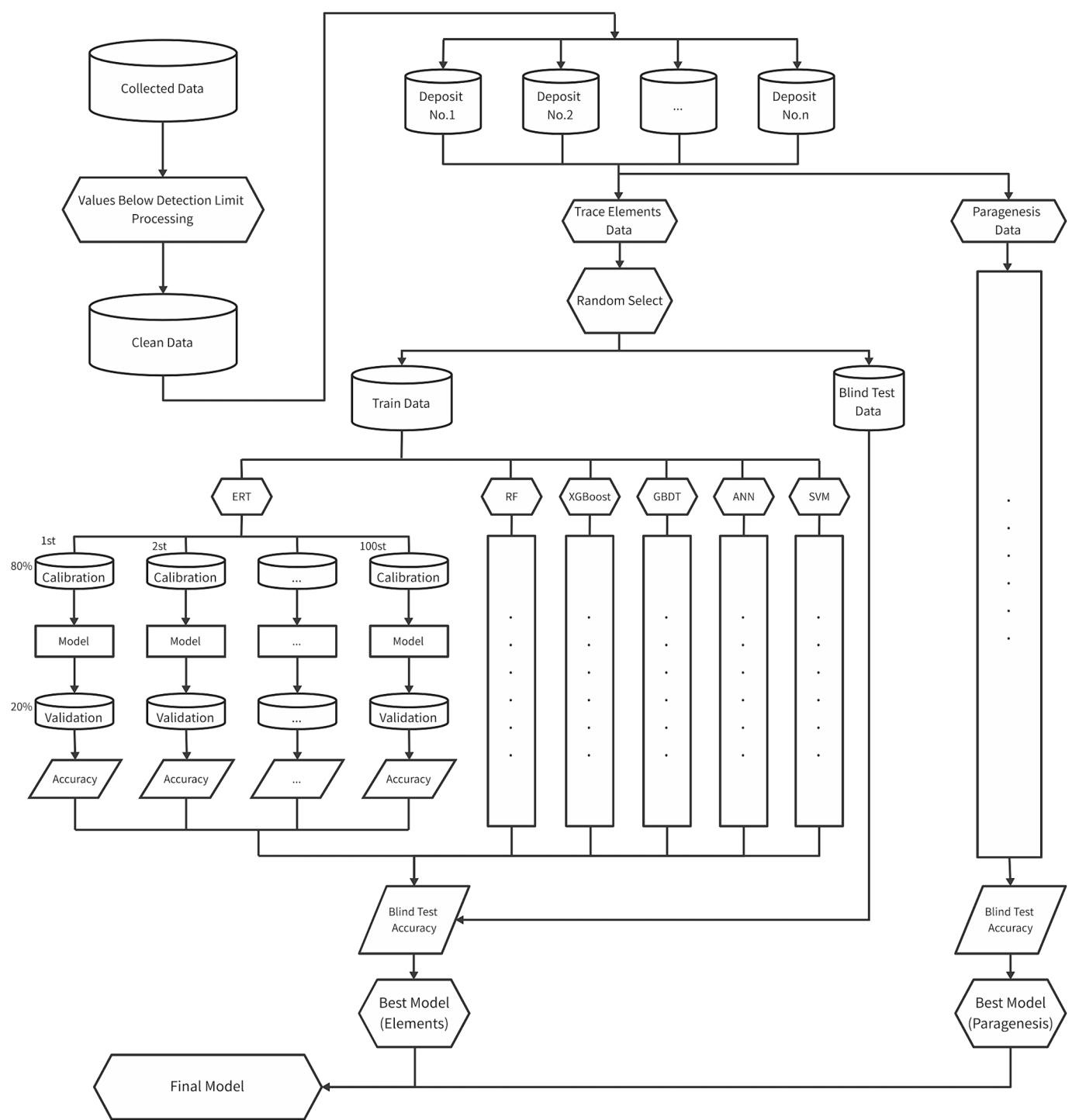


Fig. 2. Workflow chart for the development of ML algorithms in this study.

and Global Interpretability: Offers detailed explanations for individual predictions and insights into the model's overall behavior. 3) Model Agnosticism: Applicable to any machine learning model, providing a versatile tool for explanation. 4) Feature Interaction Effects: Captures both individual impacts and complex interactions between features. 5) Fair Attribution: Allocates contributions based on actual impact, ensuring equitable importance distribution. The inherent feature interpreters like RF are unable to achieve this, hence we opted for the SHAP method to interpret the model (Lundberg and Lee, 2017; Lundberg et al., 2020).

c. Composite model construction: Mineral assemblage data is

represented as a binary 0 or 1 in the dataset (0 = absence of a mineral, 1 = presence of a mineral), which is distinct from the continuous data represented by trace-element concentrations. Since ML algorithms exhibit varying performance when applied to different data types (Bishop and Nasrabadi, 2006; Gareth et al., 2013), constructing a composite model that employs different algorithms is essential for improving prediction accuracy when dealing with datasets containing diverse data types (Dietterich, 2000; Rokach, 2010). Both models can distinguish between the five deposit types, but their precision is different for the different deposit types. Specifically, the model based on trace-element chemistry of sphalerite can better distinguish MVT, skarn,

epithermal, and SEDEX deposits, but cannot recognize VMS deposits, whereas the models based on the associated mineral assemblages can better identify VMS deposits. To capture the benefits of both models, we constructed a composite ML model consisting of the two independent ML models — one model to distinguish VMS deposits from other deposit types and another model to differentiate the remaining four deposit types. We then evaluated the composite model using a four-fold cross-validation technique.

3.2. Model evaluation

To identify the most suitable ML algorithm for deposit classification using sphalerite chemistry, the quality of each model's classification was assessed based on precision, confusion matrices, and receiver operating characteristic (ROC) curves. Precision represents the probability of correct classification by the model. The confusion matrix and ROC curve serve as essential tools for evaluating multiclassification models, providing insights into model accuracy and discriminative ability, respectively (Sokolova and Lapalme, 2009). Confusion matrices are tabular representations of correct and incorrect predictions for each class; they facilitate the calculation of performance metrics, such as precision, recall, and F1-score (Provost and Fawcett, 2013), which quantify a model's classification ability and provide a basis for comparison with alternative models (Hastie et al., 2009). Conversely, ROC curves illustrate the trade-off between sensitivity and specificity by plotting the true positive rate against the false positive rate for varying classification thresholds (Fawcett, 2006). The area under the ROC curve (AUC) offers a single scalar value that summarizes a model's discriminative ability, with higher values indicating superior performance (Hand and Till, 2001). By utilizing precision, confusion matrices, and ROC curves, a comprehensive understanding of a multiclassification model's performance can be obtained, enabling informed decisions regarding model selection and optimization (Sokolova and Lapalme, 2009).

4. Results

4.1. Machine learning model

Fig. 3 illustrates the probability density distribution of classification accuracy based on two primary datasets — sphalerite chemistry (12 trace elements, each represented as individual features; Fig. 3a) and mineral assemblages (15 distinct minerals parameters as 0 or 1; Fig. 3b). The performance of the six distinct ML algorithms (XGBoost, SVM, RF, ERT, GBDT, and ANN) in classifying mineral deposits was assessed based on 100 iterations of each model. The results demonstrate that, when using sphalerite trace-element chemistry, the RF model exhibits the best

performance, accurately classifying the deposit type for 75 % of the sphalerite samples (Fig. 3a). In contrast, the ERT model exhibits the best performance when using mineral assemblage data, accurately classifying the deposit type of 71 % of the samples (Fig. 3b).

Here we will only focus on the RF and ERT models because those were most accurate; the results of other models are shown in [Supplementary Data Figs. S2–S5](#). Confusion matrices and ROC curves were constructed for each of the methods. Due to the limited applicability of F1 scores to assessing the accuracy of ML models applied to multiclassification problems, we employ the F1-micro score ($F1\text{-micro} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$) as a substitute. The F1-micro score and precision of the RF classifier using sphalerite trace-element chemistry are both 75 %. The RF classifier exhibits optimal performance in identifying MVT deposits, with a recall of 96 %. It demonstrates good performance in identifying skarn, epithermal, and SEDEX deposits, with recalls of 82 %, 79 %, and 70 %, respectively. Its performance is weakest in identifying VMS deposits, with a low recall of 17 % (Fig. 4a). The ROC curve for the RF classifier corroborates these results, with AUC values for MVT, skarn, epithermal, SEDEX, and VMS deposits of 0.98, 0.98, 0.89, 0.85, and 0.79, respectively (Fig. 4b). The F1-micro score and accuracy of the ERT classifier using mineral assemblage data are both 71 %. The ERT classifier demonstrates superior performance in identifying MVT and VMS deposits, with recalls of 94 % and 77 %, respectively. Conversely, its performance is poor when distinguishing skarn, epithermal, and SEDEX deposits, with recalls of 43 %, 47 %, and 0 %, respectively (Fig. 4c). The ROC curve shows similar results, with AUC values for MVT, skarn, epithermal, SEDEX, and VMS deposits of 0.83, 0.96, 0.59, 0.74, and 0.99, respectively (Fig. 4d). It is important to note that, although the ERT classifier exhibits a large AUC for skarn deposits, the recall is low. This observation suggests that, although the classifier is capable of differentiating between positive and negative instances effectively for this deposit type, it may miss some true positives.

In light of these results, we have constructed the composite ML model using the RF algorithm with sphalerite trace-element data and the ERT algorithm with mineral assemblage data. The RF model is designed to distinguish among MVT, skarn, epithermal, and SEDEX deposit types, whereas the ERT model is employed to identify VMS deposits. Assessment of this composite model using four-fold cross-validation (each fold has an equal proportion of different deposit types) demonstrates that it has an average precision of 85 ± 1.4 % (average $\pm 2\sigma$; 85 %, 86 %, 85 %, 84 %) based on the test dataset. In the four-fold cross-validation, the F1-micro score closely approximates the precision, with an average value of 84.8 ± 1.7 % (average $\pm 2\sigma$; 84 %, 86 %, 85 %, 84 %). The composite model excels in identifying MVT deposit types, with an average recall of 95 %. It performs well in distinguishing skarn, epithermal, and VMS deposits, with average recalls of 80 %, 81 %, and 83 %, respectively. Its

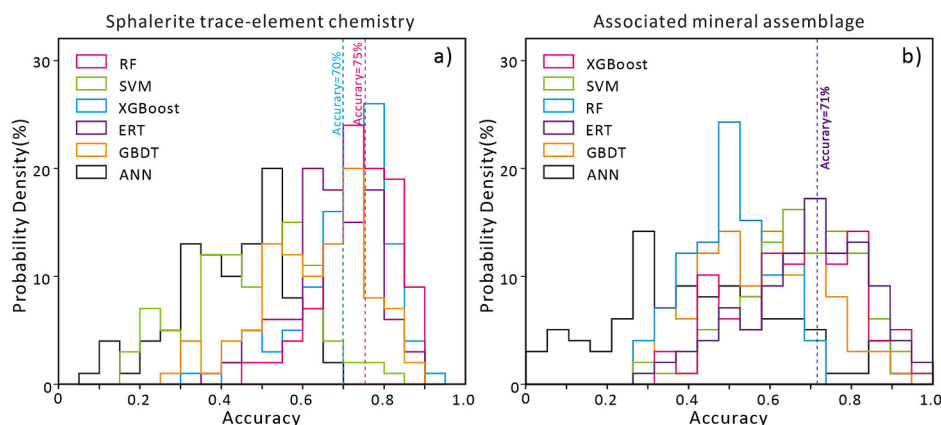


Fig. 3. Probability density distributions of the accuracy for six ML algorithms. (a) ML algorithm based on the trace-element chemistry of sphalerite (12 features) and (b) ML algorithm based on mineral assemblage data (15 features).

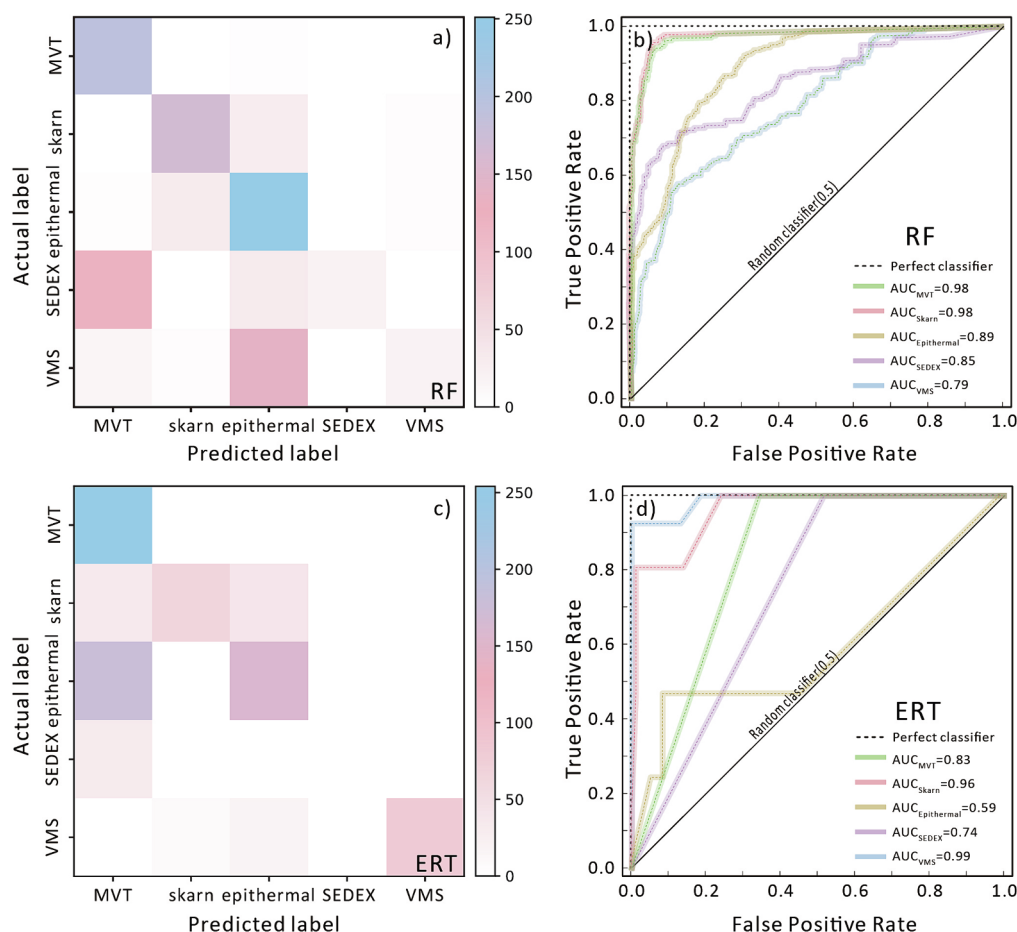


Fig. 4. (a) Confusion matrix for the RF classifier, (b) receiver operating characteristic (ROC) curves for the RF classifier, (c) confusion matrix for the ERT classifier, and (d) receiver operating characteristic (ROC) curves for the ERT classifier.

performance diminishes slightly in the case of SEDEX deposits, where the average recall is 63 % (Fig. 5). These results are similar to those obtained from the ROC curve, with average AUC values for MVT, skarn, epithermal, SEDEX, and VMS deposits of 0.98, 0.94, 0.87, 0.95, and 0.92, respectively (Fig. 6). It is important to note that, for SEDEX deposits, the higher AUC value coupled with the lower recall indicates that the model is less likely to misclassify other deposit types as SEDEX, but more likely to misclassify SEDEX deposits as other deposit types.

4.2. Evaluation of feature importance on model performance

To characterize the importance of the various trace elements and mineralogical assemblages to classifying mineral deposit types using the RF and ERT models, we utilize the SHAP algorithm, which identifies how much a given feature contributes to model prediction. SHAP values can be utilized to measure the impact of features on the output of a model, where a larger absolute SHAP value of a feature indicates a greater influence on the model's output. In the context of multi-classification problems, the larger the positive SHAP value of a feature in a given category, the more likely it will be classified into that category. The SHAP values of individual features are illustrated using a bar chart, which provides an intuitive reflection of the influence of different features on the model output (Fig. 7). The SHAP values of individual samples are illustrated using heatmaps (Fig. 8). In this figure, the vertical axis represents the sample numbers, while the horizontal axis represents the different features. The color of the bar denotes the model's output value (probability) for a particular sample. For a specific features, a significant vertical color difference, coupled with a deeper color, suggests that the feature is effective for classification, and vice

versa. This shows whether a feature is valid on the whole dataset or only on a portion of the data.

Based on sphalerite trace-element chemistry, Mn consistently has the greatest absolute SHAP value of any trace element, indicating that it has the greatest influence on deposit type classification; Ge and Co exhibit the next highest absolute SHAP values (Fig. 7a). Indium and Sn have high absolute SHAP values for SEDEX and VMS deposits, respectively, suggesting that their concentrations play an important role in classifying these deposit types (Fig. 7a). The SHAP absolute values for Cd, Fe, Ga, Pb, Sb, Ag, and Cu are low (Fig. 7a), and they do not demonstrate a pervasive influence across the entire dataset (Fig. 8), indicating that they have minimal impact on the classification.

Based on mineral assemblage data, chalcopyrite has the highest absolute SHAP value and, therefore, exerts the strongest influence on distinguishing among most of the deposit types, except for VMS, followed by tetrahedrite, magnetite, arsenopyrite, tellurides, and pyroxene (Fig. 7b, 8). Importantly, the high absolute SHAP value for magnetite in skarn deposits illustrates that the presence of magnetite is critical to identifying skarn deposits (Fig. 7b). Similarly, the presence of arsenopyrite and tellurides are important for classification of VMS deposits given their high absolute SHAP values for this deposit type (Fig. 7b). Lastly, pyroxene is important to distinguish between MVT and Skarn deposits.

Beyond analyzing average absolute SHAP values, another perspective to consider is the sign of the SHAP values, which can reveal whether a feature is positively or negatively correlated with a specific deposit type. To do this, a series of scatter plots of SHAP values are plotted for each category (Figs. 9, 10). The variables are arranged in order of importance in distinguishing a given deposit type; the color of the data

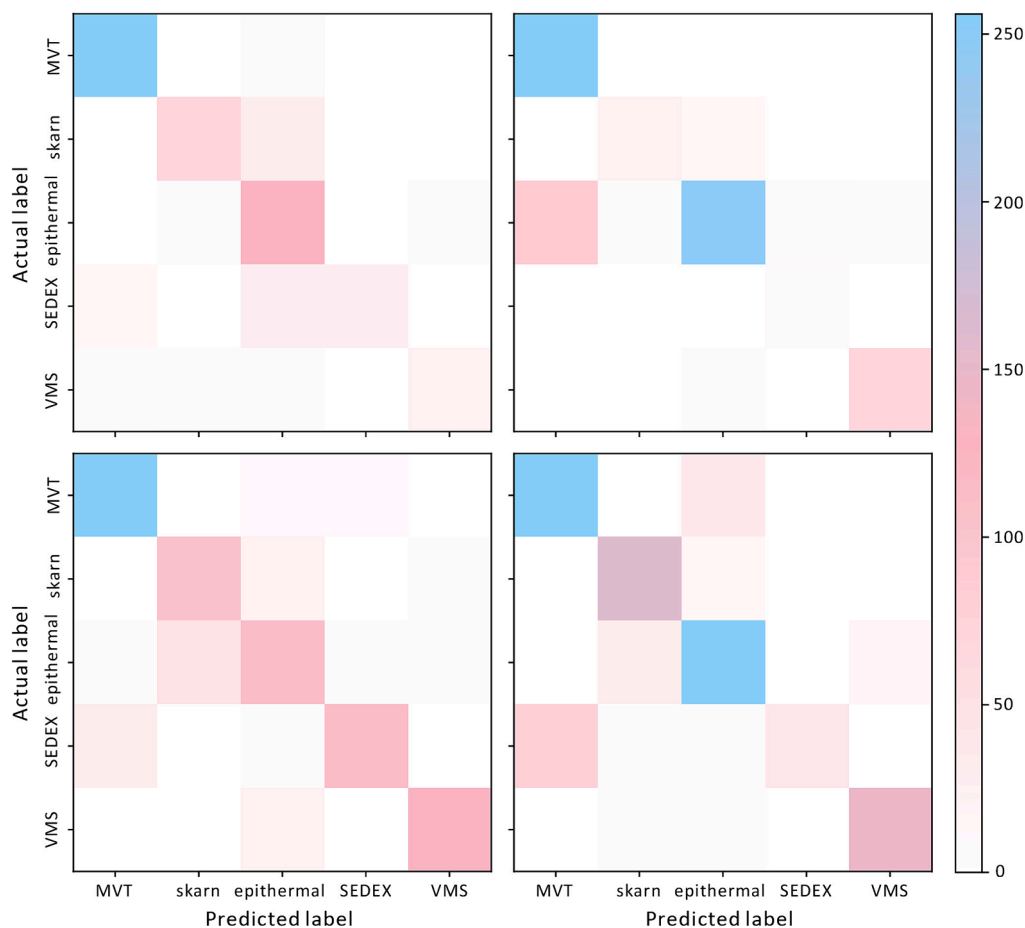


Fig. 5. Confusion matrices showing the cross-validation results for the composite model. (a) First cross-validation (accuracy = 85 %), (b) second cross-validation (accuracy = 86 %), (c) third cross-validation (accuracy = 85 %), and (d) fourth cross-validation (accuracy = 84 %).

points (blue to red) represents the feature value. Furthermore, for a specific feature, a greater cluster of datapoints at high absolute SHAP values signifies a greater importance of that feature to the model output, whereas a greater cluster of datapoints with values closer to 0 signify a lower impact. These plots clearly illustrate how each variable influences the classification results (Figs. 9, 10). For trace elements, the criteria by which the model classifies the input as a specific deposit type are: epithermal = high Mn + high Sn + low Co, Ge (Fig. 9a); MVT = low Mn + high Ge (Fig. 9b); SEDEX = low Mn + low Ge (Fig. 9c); skarn = high Co + high Mn (Fig. 9d); VMS = high Mn + low Sn + high Cd (Fig. 9e). For the mineral assemblage data, the criteria by which the model classifies the input as a specific deposit type are: epithermal = high chalcopyrite + high tetrahedrite + low magnetite + low arsenopyrite (Fig. 10a); MVT = low chalcopyrite + low tetrahedrite + low pyroxene + low telluride (Fig. 10b); SEDEX = low chalcopyrite + low tetrahedrite + low telluride + low pyroxene (Fig. 10c); skarn = high magnetite + high chalcopyrite (Fig. 10d); VMS = high arsenopyrite + high telluride (Fig. 10e).

5. Discussion

5.1. Significance of trace elements in sphalerite to deposit classification

The concentrations of Mn, Ge, Co, Sn, and Cd in sphalerite are highly variable depending on the nature of the deposit, with notable differences exhibited by MVT, skarn, epithermal, SEDEX, and VMS deposits. These geochemical variations correspond to their unique formation conditions and geochemical affinities (Guilbert and Park Jr, 2007).

The models suggest that sphalerite from MVT deposits contains high

Ge and low Mn, a finding that is consistent with natural examples. MVT deposits commonly form from low-temperature (100–200 °C), high-salinity brines, and tend to yield sphalerite with lower Mn and higher Ge concentrations (Leach et al., 2005). This can be attributed to the fact that Mn is typically enriched in magmatic settings (Frenzel et al., 2016) and its solubility is significantly reduced in low-temperature conditions (Leach et al., 2005; Cook et al., 2009). In contrast, Ge is typically associated with low-temperature environments (Cook et al., 2009). The models suggest that sphalerite from skarn deposits have characteristically high Co and Mn concentrations, which is consistent with natural examples. This reflects the high-temperature (400–600 °C) and metal-rich nature of the magmatic fluids from which they precipitate (Einaudi et al., 1981; Meinert, 1992; Cook et al., 2009), and the fact that higher temperatures favor the substitution of Mn into sphalerite (Maslennikov et al., 2009; Lockington et al., 2014; Maslennikov et al., 2017). In addition, skarns are typically enriched in Co (Zhao et al., 2019), suggesting that the mineralizing fluids are rich in Co, as well as the minerals that precipitate from them, such as sphalerite. This Co enrichment is supported by the fact that some skarn deposits are genetically associated with mafic-ultramafic rocks, which often have elevated Co contents (Gunn, 2014; Maslennikov et al., 2017; Zhao et al., 2019). The models suggest that sphalerite in epithermal deposits contains high Mn and Sn, but low Co and Ge. These geochemical patterns reflect the high temperature (100–500 °C) and low sulfur fugacity hydrothermal conditions typically associated with epithermal systems (Shimizu and Shikazono, 1985; Hedenquist et al., 2000; Einaudi et al., 2005; Reich et al., 2005). The models suggest that SEDEX deposits contain sphalerite with low Mn and Ge as a result of their lower formation from low-temperature (<150 °C) basinal brines (Deb and Goodfellow, 2004). The models suggest that

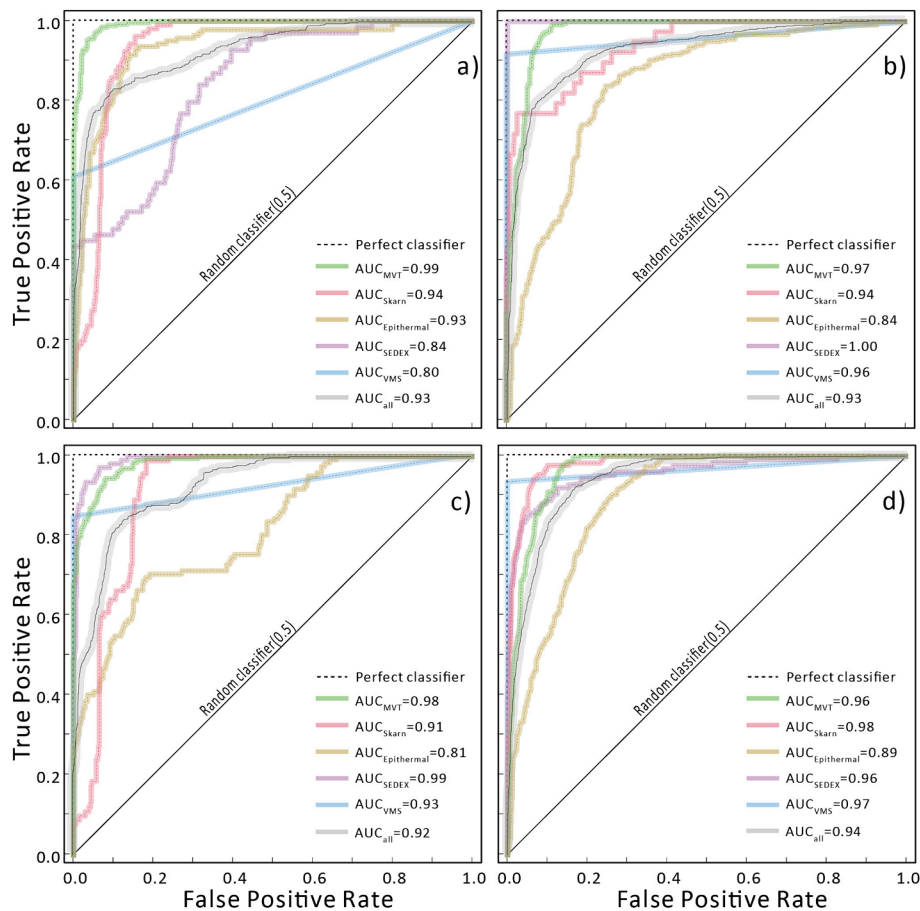


Fig. 6. Receiver operating characteristic (ROC) curves for the composite model. (a) First cross-validation (AUC = 0.93), (b) second cross-validation (AUC = 0.93), (c) third cross-validation (AUC = 0.92), and (d) fourth cross-validation (AUC = 0.94).

sphalerite in VMS deposits contains high Mn and Cd, but low Sn as a result of the high-temperature (250–350 °C) and low sulfur fugacity conditions involved in their formation (Large, 1992; Franklin et al., 2005; Galley et al., 2007).

5.2. Significance of mineral assemblages to deposit classification

Different geological conditions tend to form distinct mineral assemblages, with factors such as fluid composition, temperature, and pressure playing pivotal roles in this process (Guilbert and Park Jr, 2007). While this non-numeric information often carries a slight element of subjectivity, it can still provide insights beyond trace elements that significantly aid in identifying deposit types. Prior to discussing mineral assemblages, it is worth noting that the ERT model, based on mineral assemblages, only exhibits satisfactory classification accuracy for MVT, skarn, and VMS deposits. Therefore, we will focus on these three deposit types.

Consistent with natural examples, the models suggest that the mineral assemblage of MVT deposits is remarkably simple, seldom featuring minerals other than sphalerite, galena, and pyrite. This can be attributed to several features. First, the alteration processes inherent to MVT deposits are simple (Ridley, 2013). This is primarily attributable to the predominantly low-temperature, epigenetic mineralization processes that govern MVT deposits, which typically occur in sedimentary basins under relatively stable hydrothermal conditions (Leach et al., 2005). Since the complexity of alteration processes often corresponds to the complexity of the resultant mineral assemblages (Rose, 1979), the simple alteration processes in MVT deposits results in a simple mineral assemblage. Second, the materials required to form MVT deposits often

originate from a single source. It is widely agreed that metals are sourced from the interaction of basinal brines with sedimentary rocks (Leach et al., 2005). The basinal brines, often sourced from or modified by evaporated seawater, migrate through the sedimentary basin and leach Zn–Pb from the host rocks (Sverjensky, 1984). The sulfur is commonly thought to originate from the reduction of sulfate in the brines by organic matter or other reduced carbon species in the host rocks (Ohmoto et al., 1979). Third, MVT deposits form over relatively short geological timescales. MVT deposits often form in response to tectonic events, such as basin inversion or orogeny (Leach et al., 2005). These tectonic events drive fluid flow, creating pressure gradients that can result in the rapid migration of brines through permeable strata. The process of metal leaching, transportation, and deposition can, therefore, occur over relatively short geological timescales, possibly ranging from thousands to hundreds of thousands of years (Symons et al., 2009).

Consistent with natural examples, the models suggest that the mineral assemblage of skarn deposits contains sphalerite, galena, pyrite, chalcopyrite and magnetite. In addition to sphalerite, galena, and pyrite, significant amounts of chalcopyrite and magnetite are also typically present. This is due to the high concentrations of Cu in and the elevated temperatures (400–600 °C; Einaudi et al., 1981) of the hydrothermal fluids associated with skarns. Copper is highly soluble in magmatic fluids, particularly in the presence of chlorine and sulfur ligands (Williams-Jones and Heinrich, 2005). Furthermore, Cu can be leached from the cooling magma body during the late stages of a hydrothermal system. These factors contribute to the high Cu concentration of hydrothermal fluids associated with skarn deposits (Sillitoe, 2010) and promote the precipitation of chalcopyrite. On the other side, the presence of significant amounts of magnetite indicates the high temperature.

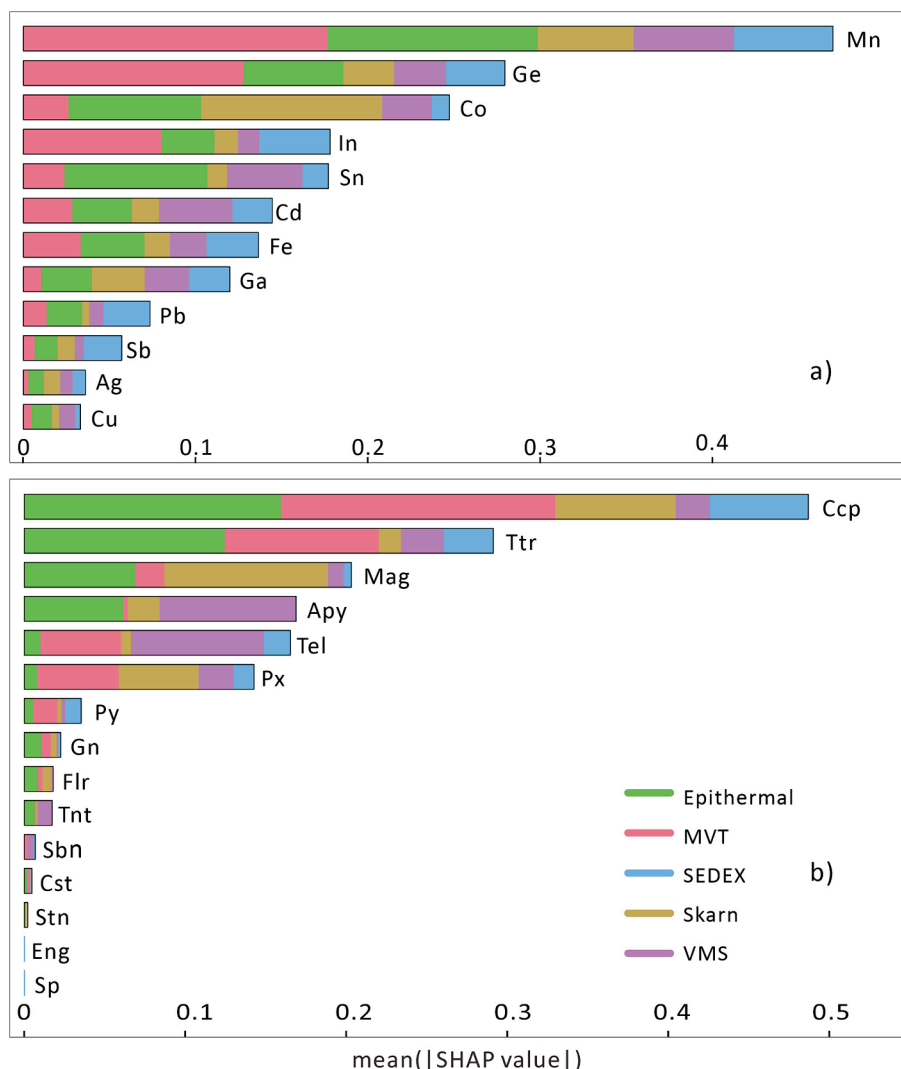


Fig. 7. Stacked bar plots illustrating the feature importance of (a) the RF classifier based on sphalerite geochemistry and (b) the ERT classifier based on mineral assemblage data. Sp = Sphalerite; Gn = Galena; Py = Pyrite; Ccp = Chalcopyrite; Ttr = Tetrahedrite; Tel = Tellurides; Mag = Magnetite; Stn = Stannite; Flr = Fluorite; Cst = Cassiterite; Eng = Enargite; Sbn = Stibnite; Apy = Arsenopyrite; Tnt = Tennantite; Px = Pyroxene.

Because the precipitation of magnetite can serve as an indicator of high-temperature fluids ($>400\text{ }^{\circ}\text{C}$), commonly associated with skarn, porphyry, or iron-oxide copper-gold (IOCG) deposit types (Barton and Johnson, 1996).

The models suggest that in VMS deposits, apart from sphalerite, galena, and pyrite, a distinct feature is the presence of significant amounts of arsenopyrite and tellurides. This can be attributed to the high concentrations of As and Te in the moderate-high-temperature, volcanic-associated hydrothermal fluids ($250\text{--}350\text{ }^{\circ}\text{C}$; Franklin et al., 2005). Furthermore, arsenopyrite generally precipitates from hydrothermal fluids at moderate to high temperatures, typically between $200\text{--}350\text{ }^{\circ}\text{C}$ (Fleet and Mumin, 1997), which is similar to the temperatures at which VMS deposits form. Although the concentration of Te in the crust is relatively low, it is concentrated during the later stages of crystallization of magma and can be mobilized by volatile-rich phases (Fleet et al., 1993), eventually precipitating as tellurides at moderate-high temperatures ($>200\text{ }^{\circ}\text{C}$; Saunders, 1986).

5.3. Practical applications

To assess the real-world applicability of the combined ML model to distinguish among deposit types, we use the model to classify the Jinding deposit. The Jinding sediment-hosted Zn–Pb deposit in western

Yunnan (Fig. 1), China, is the fourth largest Zn deposit in Asia (200 Mt of ore containing 6.1 % Zn and 1.3 % Pb; Xue et al., 2007). Some remarkable aspects of the Jinding deposit include the extremely complex and compositionally variable breccias, abundant gypsum, and the unusual textures of the sandstone-hosted ores (Leach et al., 2017), which is unusual for MVT deposits. The mineral assemblage within the Jinding deposit is simple, primarily comprising sphalerite, galena, pyrite, and hematite (Li et al., 2021). The Jinding deposit has been classified as syngenetic, SEDEX, and MVT deposits, as well as a unique ‘Jinding-type’ deposit (Xue et al., 2007). However, recent work has presented compelling evidence that the ore-hosting sandstones are specifically associated with the formation of the Jinding diapir. In this model, the sandstones in the Jinding deposit represent a melange of material extruded from the diapiric zone onto the surface that mixed with sediment from a local fluvial system (Leach et al., 2017). Accordingly, the Jinding deposit should be classified as MVT sub-type, which is hosted in a diapiric environment (Leach et al., 2017).

To classify the Jinding deposit using our combined ML model, we input the average concentrations of trace elements in sphalerite and the associated mineral assemblages from this deposit (Ye et al., 2011) into the model, then use a force plot, which combines the SHAP values and input data, to illustrate the prediction process (Fig. 11). In the figure, the ‘base value’ represents the initial parameters of the trained model, ‘f(x)’

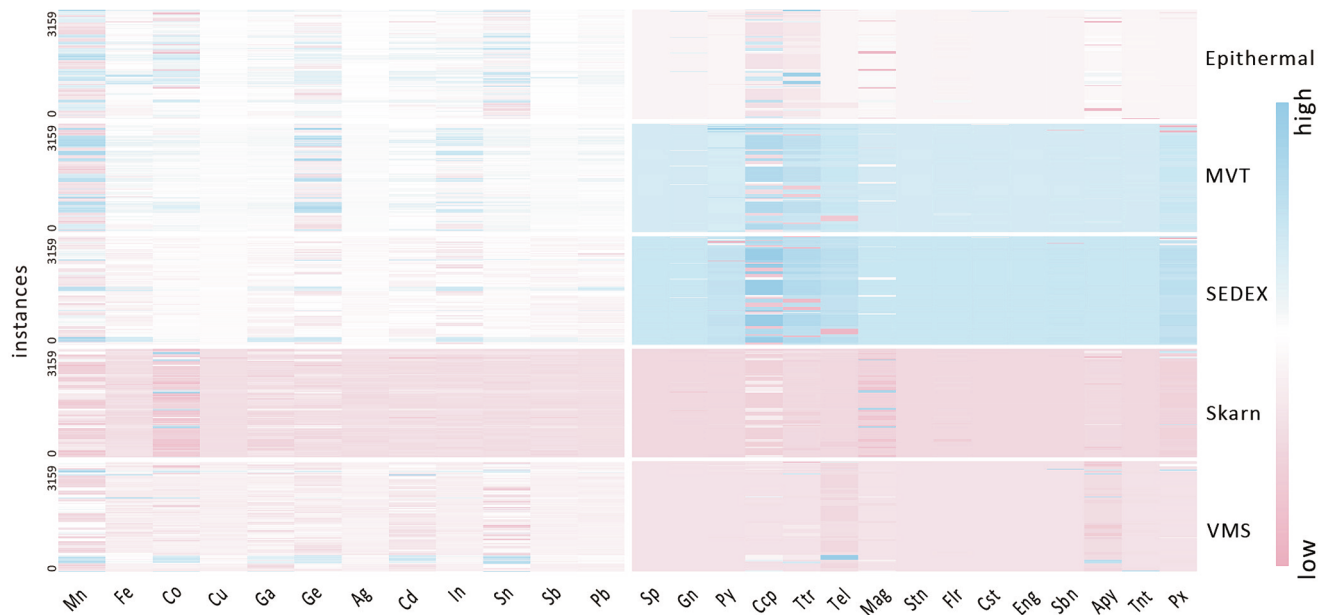


Fig. 8. Heat map of SHAP values for deposit types obtained from the RF and ERT classifiers.

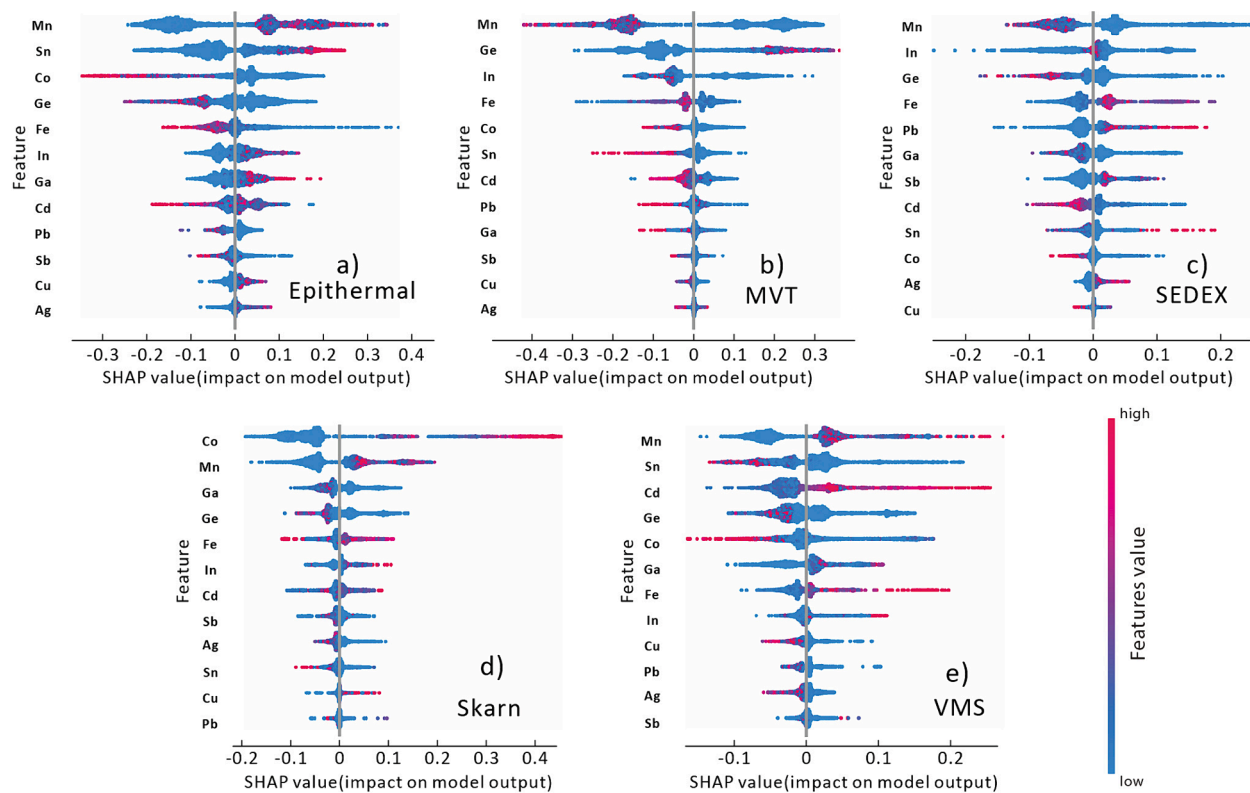


Fig. 9. Binary diagrams summarizing the SHAP values from the RF classifier for (a) epithermal, (b) MVT, (c) SEDEX, (d) skarn, and (e) VMS deposits.

is the model's output (classification possibility), where a higher value suggests a greater likelihood of the given category, and the red and blue blocks represent variables that contributed positively and negatively to the predicted value, respectively, with longer block lengths indicating a greater contribution to the prediction. The results indicate that, based on trace-element chemistry of sphalerite, the Jinding deposit is most likely a MVT deposit as it has a $f(x)$ value of 0.8, which is significantly higher than the $f(x)$ value of the other deposit types (Fig. 11a–e). Furthermore, the predicted value of the ERT model based on the associated mineral

assemblage, which is designed to distinguish VMS deposits, is 0.01, effectively ruling out the possibility that Jinding is a VMS deposit (Fig. 11f). The model classification is consistent with the results of Leach et al. (2017), demonstrating the high performance of the model even in geologically complex scenarios.

5.4. Model superiority

Although machine learning models have been previously developed

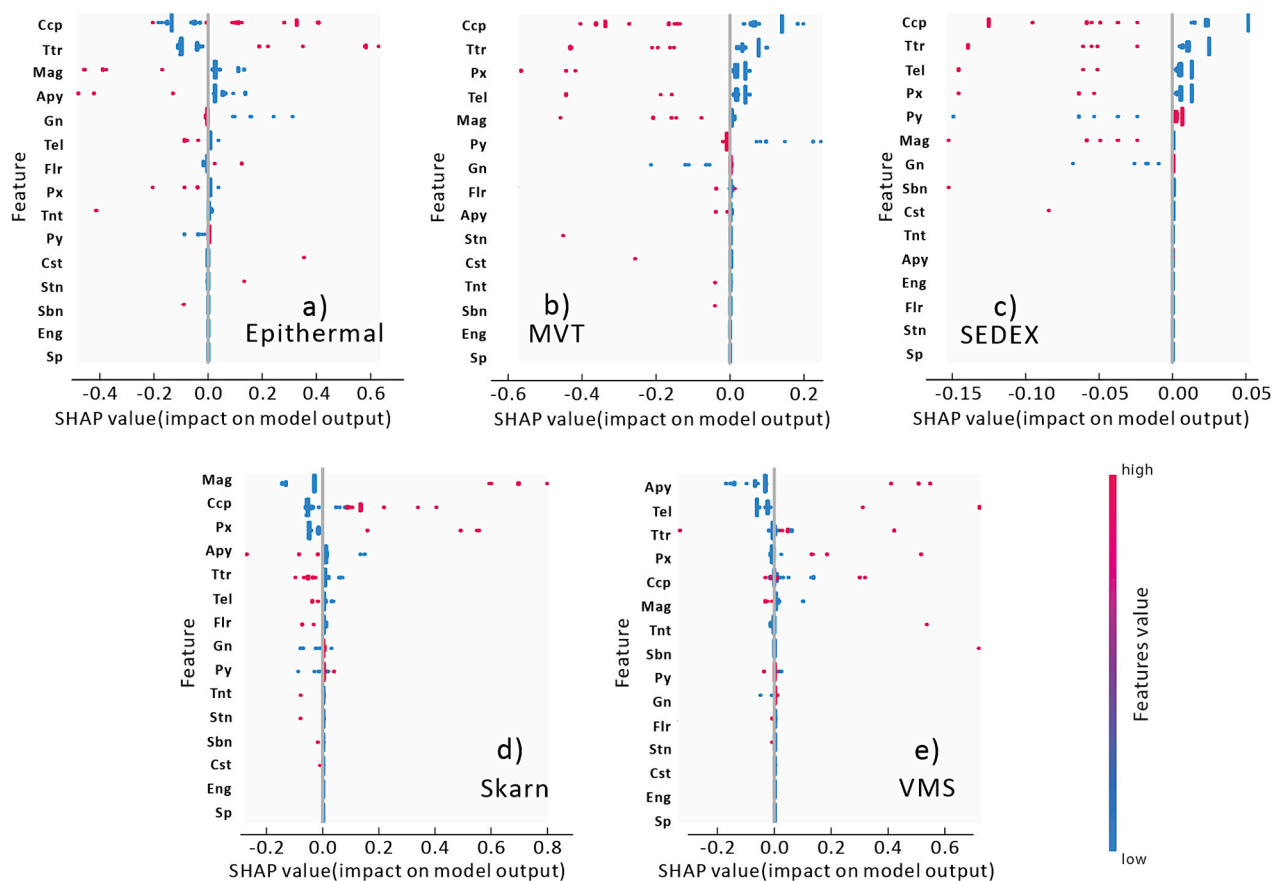


Fig. 10. Binary diagrams summarizing the SHAP values from the ERT classifier for (a) epithermal, (b) MVT, (c) SEDEX, (d) skarn, and (e) VMS deposits. Sp = Sphalerite; Gn = Galena; Py = Pyrite; Ccp = Chalcopyrite; Ttr = Tetrahedrite; Tel = Tellurides; Mag = Magnetite; Stn = Stannite; Flr = Fluorite; Cst = Cassiterite; Eng = Enargite; Sbn = Stibnite; Apy = Arsenopyrite; Tnt = Tennantite; Px = Pyroxene.

to distinguish deposit types based on sphalerite chemistry, there are significant limitations in their applicability. Sun and Zhou (2022) attempted to classify Pb–Zn-rich deposit types using sphalerite chemistry and various machine learning algorithms (DT, KNN, NB, RF, and SVM), but the training and test sets were separated based on sphalerite trace-element chemistry rather than by deposit. This would have resulted in some of the deposits in the test set having already appeared in the training set, leading to severe model overfitting. Li et al. (2023) differentiated among different deposit types using sphalerite trace-element chemistry and the RF algorithm, establishing a training set and test set separated based on deposit type, thereby addressing the overfitting issue. However, VMS deposits could not be distinguished using only sphalerite trace-element chemistry, and the quality of the model results was affected by the significant imbalance in population size for each mineral deposit type in the test set.

Our model addresses these issues in two ways. First, our model considers the trace-element content of sphalerite in a single deposit as a unit, establishing training and test sets that effectively mitigate overfitting. Second, our model parameterizes and incorporates mineral assemblage information, which resolves the issue of classifying VMS deposits. Furthermore, by employing the SHAP method to interpret the machine learning model, we have intuitively demonstrated the impact of various variables on the model's output. We elaborately demonstrate the model's prediction process, substantially addressing the 'black box' issue associated with machine learning. By introducing weights, we mitigated the potential impact of uneven sample distribution on model performance, thereby enhancing the model's generalizability. Taken together, this study represents a prime example of how ML can be appropriately leveraged to address geological issues. It is worth noting that the application of machine learning models requires careful

consideration and should not be extended to new types of deposits for which they have not been trained.

6. Conclusions

- (1) The Random Forest algorithm is the most suitable for classifying deposit types using sphalerite trace-element chemistry, whereas the Extra Random Trees algorithm proved to be the most effective when using mineral assemblage data.
- (2) Manganese (Mn), Ge, Co, In, and Sn are key elements, and chalcopyrite, tetrahedrite, magnetite, and arsenopyrite are key minerals to discriminate among different deposit types. We highlight the importance of utilizing both data types in combination to accurately classify deposit types.
- (3) The composite ML model that incorporates sphalerite trace-element chemistry and mineral assemblage data demonstrates verifiably high accuracy ($85 \pm 1.4\%$) and reliability ($AUC = 0.93 \pm 0.01$).
- (4) This study provides an executable program of the classifier (preserved at <https://doi.org/10.5281/zenodo.8103535>), enabling easy access to the combined ML model, allowing it to be readily applied to mineral exploration and prospecting.

7. Implications

With the advent of artificial intelligence (AI) and its rapid advancement, an increasing number of disciplines are leveraging this technology to address complex problems. However, despite its transformative potential, the efficacy of ML can be compromised by improper application. It is paramount to apply rigorous scientific experimental

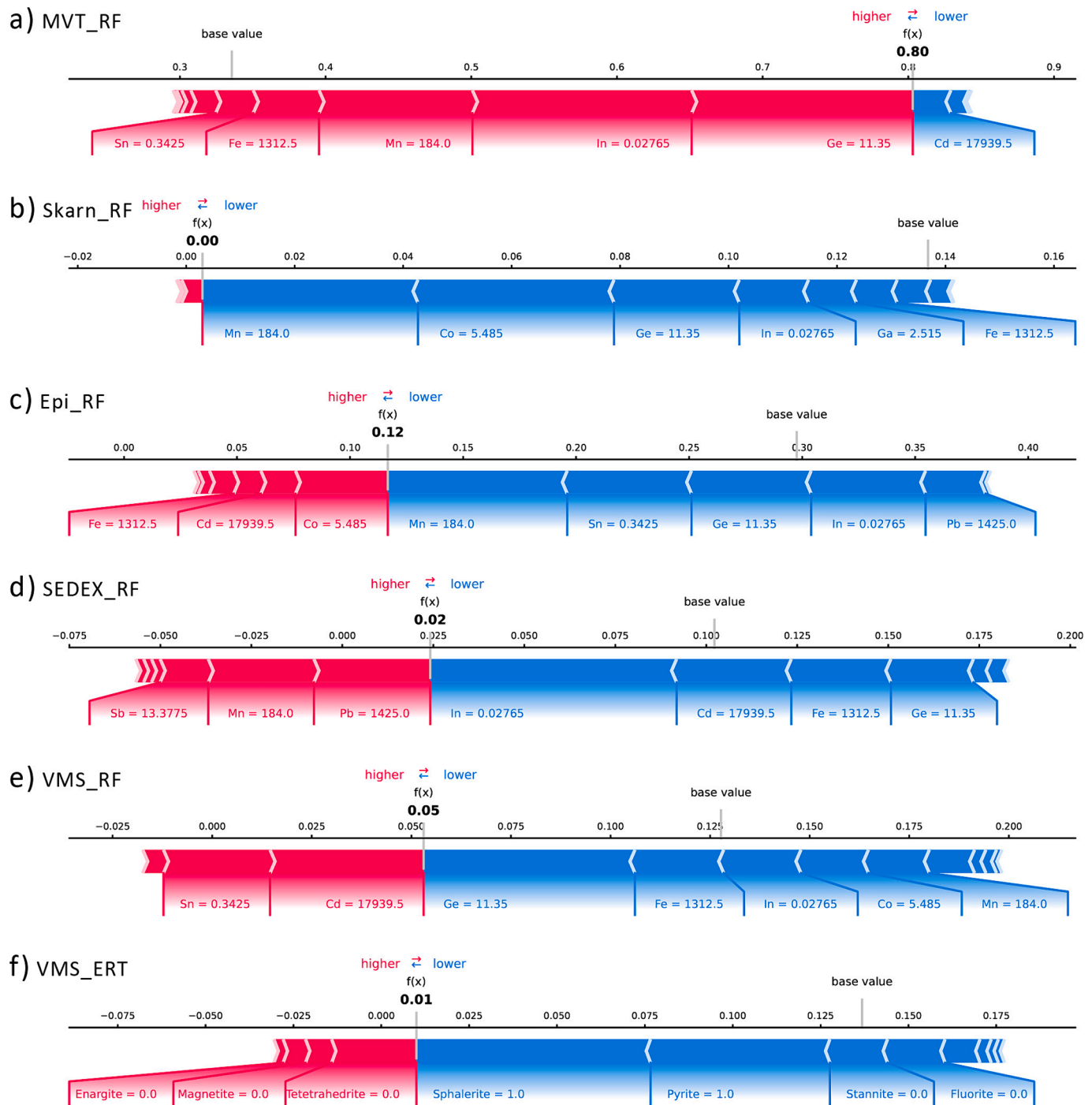


Fig. 11. Force plot for composite model predictions for the Jinding deposit.

design and evaluation methods to ensure the validity of the outcomes; otherwise, such approaches may yield biased or even erroneous results. Notably, current applications of ML in geosciences predominantly concentrate on geochemical data, often overlooking non-numerical geological information, such as mineralogy and mineral textures. This oversight potentially limits the breadth and accuracy of the interpretations. To enrich our understanding and to solve more intricate practical issues, it is imperative to incorporate non-numerical geological information into ML applications in the geosciences.

Data availability statement

The dataset and Version 1.0 of the software Sp.Type.exe is preserved at <https://doi.org/10.5281/zenodo.8103535>, and available for download. Further descriptions regarding the construction of the machine learning models are available through the Keras (<https://keras.io/api/>) and scikit-learn (<https://scikit-learn.org/stable/index.html>) documentation.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.oregeorev.2024.106076>.

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